Project Design Phase-I
Problem – Solution Fit Template

Date	8 Mar 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The rapid growth of Electric Vehicles (EVs) has created new challenges in understanding charging patterns, battery performance, and vehicle efficiency. Large amounts of EV data such as charging time, power consumption, driving range, battery health, and vehicle specifications are available, but they are often scattered across multiple sources and difficult to analyze.

City planners, EV fleet managers, and potential EV buyers need a clear way to interpret this data in order to make better decisions. However, without proper visualization tools, identifying trends, comparing vehicle performance, and understanding charging behavior becomes complex and time-consuming.

To address this problem, a **Tableau-based visualization tool** is developed to transform raw EV datasets into interactive dashboards and data stories. The system collects EV data, stores it in a database, and connects it with Tableau to generate visual insights such as charging station distribution, vehicle efficiency, price comparisons, and battery performance metrics.

The solution enables users to explore EV data through **maps, charts, and interactive dashboards**, helping them identify patterns, compare EV models, and analyze charging infrastructure demand.

This approach helps different stakeholders:

- **City Energy Planners** can analyze EV charging patterns and plan new charging infrastructure.
- **Fleet Managers** can monitor battery performance and vehicle efficiency to reduce operational costs.
- **Consumers** can compare EV models based on range, price, and efficiency before making a purchase decision.

By integrating **data analytics, visualization, dashboards, and web integration using Flask**, the proposed solution provides an efficient platform for analyzing EV charging and performance data. This improves decision-making, enhances data accessibility, and supports the growing EV ecosystem.

Define CS, fit into CC

Explore AS, differen

Focus on J&P, tap into BE, understand RC

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1. CUSTOMER SEGMENT(S) CS

Who is your customer?

Adults who can drive vehicles and possess a valid driving license..

6. CUSTOMER CONSTRAINTS

Limited information about EV range and performance
Lack of clear visualization tools
Limited awareness of charging station locations
Difficulty comparing different EV models

5. AVAILABLE SOLUTIONS

Which solutions are available to the customers when they face the problem

Customers currently use:

- Online EV websites and comparison platforms
- Navigation apps to locate charging stations
- Manufacturer websites for EV specifications

Pros: Easy access to information, wide data availability.
Cons: Data is scattered, limited visualization, difficult to compare EV range and charging infrastructure.

2. JOBS-TO-BE-DONE / PROBLEMS

- Compare electric vehicle range, price, and performance.
- Find nearby EV charging stations.
- Understand EV battery efficiency and driving range.

9. PROBLEM ROOT CAUSE RC

EV data is scattered across multiple platforms.

Lack of clear visualization tools for EV analysis.

Limited awareness

7. BEHAVIOUR BE

Searches online for EV specifications and comparisons

Uses navigation apps to locate charging stations

Reads reviews and EV market information

Compares different EV models before making decisions

3. TRIGGERS TR

Rising fuel prices

Government incentives for electric vehicles

Increasing environmental awareness

News and advertisements about EV technology

Growth of EV charging infrastructure

10. YOUR SOLUTION SL

An interactive visualization tool developed using Tableau that analyzes electric vehicle data. It provides dashboards to compare EV range, price, battery efficiency, and shows EV charging station locations, helping users easily understand EV performance and infrastructure.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

Search EV specifications and comparisons on websites
Use navigation apps to find charging stations
Read EV reviews and articles

8.2 8.2 OFFLINE

Visit EV dealerships for vehicle information
Discuss EV options with friends or experts
Observe charging stations in public places

4. EMOTIONS: BEFORE / AFTER EM

Before: Confused, uncertain, worried about EV range and charging availability.

After: Confident, informed, and comfortable making EV decisions.